

Phase 2 (Parity + Reach) — Work Breakdown: phases → tasks → subtasks

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This document is the executable spine of HelixVPN **Phase 2 — Parity + Reach**: the additive evolution that takes the Phase-1 self-hostable MVP to **full Mullvad feature parity and beyond on reach**. It decomposes the work into stable, DB-ready items (HVPN-P2-NNN), each carrying *description · depends-on · deliverable · acceptance · effort · required-test-types*, grouped by capability epic. It is **spec-only** — it describes *what to build and how "done" is proven*, never the product itself (2–3 refinement passes follow).

Phase-2 discipline (the load-bearing premise [04_P2 §0/§7]): *everything is additive*. Phase 1 drew the seams — the Transport trait, Peer.endpoint candidate list, federated Coordinator, TransportPolicy.order, the NetworkMap delta protocol, PKI hooks — precisely so no Phase-2 capability reshapes an existing interface. New transports are *more impls of the same trait*; NAT traversal is *fields the map already reserved*; HA is *topology, same images*.

Authoritative companions (referenced, never re-defined): data plane 01-data-plane.md [01]; control plane DDL/protobuf/events 02-control-plane.md [02]; client core + UI 03-client-core-and-ui.md [03]; security/privacy/PKI 04-security-privacy-pki.md [04]; product scope 00-product-scope-and-principles.md [00]; the Phase-1 WBS 07-phase1-mvp-wbs.md [07]. Primary research: [04_P2] (HelixVPN-Phase2-Parity.md), [04_ARCH], [04_P0], [04_P1], [11_MST], [07_GMI], [10_KMI], [SYNTHESIS], plus the external-fact anchors [research-masque], [research-hysteria2], [research-daitya_test], [research-pki_pq_nat].

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0. How to read this WBS

Granularity. Three levels: **Epic** (HVPN-P2-Enn, a capability workstream) → **Task** (HVPN-P2-NNN, a shippable unit, ~1–10 person-days) → **Subtask** (HVPN-P2-NNN.k, a single PR-sized change). Every **leaf** (a task with no subtasks, or a subtask) is a workable item per §11.4.93 and is the unit that lands in the SQLite single-source-of-truth DB (§3).

ID scheme (stable, append-only, §11.4.54). HVPN-P2- prefix; the numeric block encodes the epic: Enn → tasks nn0...nn9, e.g. E22 (NAT traversal) → 220...229. IDs are **never renumbered, reused, or decremented**; new work appends at the next free number in the block (gaps allowed). Phase-1 ids (HVPN-P1-*) are a disjoint namespace; cross-phase deps cite the full id.

Field contract (every leaf).

Field	Meaning
Desc	What the item builds, in one or two sentences (§11.4.91 ≥6 words / ≥40 chars).
Deps	Hard upstream item IDs (must be complete first). – = none.
Deliverable	The concrete artefact(s) — file paths, crates, generated clients, manifests.
Acceptance	The falsifiable PASS condition, captured-evidence per §11.4.5/.69/.107.

Field	Meaning
Effort	T-shirt + person-days: XS(1) S(2–3) M(4–6) L(7–10) XL(11–15).
Tests	Required test types from the §1 closed vocabulary (§11.4.169).

Risk ordering (§11.4.132). Within and across epics the *highest-risk*, *hardest-to-reverse* capabilities lead: NAT traversal correctness (E22), DAITA efficacy (E21), and PQ downgrade-safety (E24) are sequenced before convenience surfaces (GitOps polish, desktop split-tunnel UI). Traffic-analysis defense and PQ negotiation are the two "subtle, easy to bluff" features — each carries a *measure-don't-assert* acceptance and a self-validated analyzer (§11.4.107(10)).

Anti-bluff (§11.4 / §11.4.27 / §11.4.107). Every Acceptance line is a *captured-evidence* assertion, not "code compiles". A leaf is complete only when its required-test-types are green **with** an evidence path in the item's `test_diary` (§11.4.149) and, where it has a user-visible surface, a window-scoped MP4 vision-verified per §11.4.159/.163. Two Phase-2 features are *privacy claims* that MUST be measured, never asserted: DAITA fingerprinting reduction (§7 / [research-daita_test]) and PQ session protection (§10). A green pass that asserts the claim without the measurement is a §11.4 PASS-bluff.

1. Required-test-types vocabulary (§11.4.169)

§11.4.169 mandates that **every** workable item declare the closed set of test types it must pass before closure; the universe is the §11.4.27 enumeration. The closed vocabulary + canonical abbreviations used in this WBS:

Abbrev	Type	Phase-2 floor expectation
UNIT	Unit	pure logic (transport framing, ladder FSM, KEM combine, hop-chain builder); mocks allowed only here (§11.4.27).
INT	Integration	real Postgres/Patroni + NATS + Redis via <code>containers submodule testcontainers</code> (§11.4.76); no mocks.
E2E	End-to-end	<code>netns rig [04_P0 §3]</code> with simulated NATs/DPI, fed by the real federated control plane.
FA	Full-automation	self-driving, re-runnable <code>-count=3</code> , no human in the loop (§11.4.98).
SEC	Security	downgrade-safety, hop-isolation capture, relay can't read traffic, mTLS, no-PQ-only, key-never-leaves (§04).
CHAOS	Chaos	kill a coordinator / PG primary / a whole region; NATS partition; mid-rekey SIGKILL (§11.4.85).
STRESS	Stress	hole-punch storms, 10k concurrent streams, relay saturation, boundary NAT timings (§11.4.85).
PERF	Performance	failover-reconnect p99, relay→direct upgrade latency, DAITA overhead %, cross-region p99 (§2 SLOs).

Abbrev	Type	Phase-2 floor expectation
BENCH	Benchmarking	per-transport goodput/CPU-per-Gbps, KEM handshake cost, relay forwarding throughput [04_P0 §8].
SCALE	Scaling	N regions × M agents soak; NATS consumer lag + PG replication lag bounded over 24 h.
DDOS	DDoS	re-armed in Phase 2 (public multi-region surface): STUN endpoint + edge handshake flood, amplification-refusal (§04).
UI	UI	widget + golden tests (path indicator, multi-hop picker, DAITA toggle, PQ chip) on the three flavors.
UX	UX	flow walkthrough (P2P upgrade, multi-hop select, region failover) with window-scoped MP4 + vision verdict (§11.4.159).
REC	Recorded-evidence	window-scoped MP4 (§11.4.154/.155) + media-validation pipeline verdict (§11.4.163).
CHAL	Challenge / HelixQA	a challenges / helix_qa bank entry scoring PASS only on captured evidence (§11.4.27/.107).

Every task lists only the **required** subset; absent types are out-of-scope for that item by design, not omission. Unlike Phase 1, DDOS is now **applicable** — the multi-region public edge + STUN endpoints are an internet-facing surface (§11.4.6 — the Phase-1 NOT_APPLICABLE: single-node-selfhost deferral is re-armed here).

2. Phase-2 Definition-of-Done — the 8 acceptance gates + 5 SLOs

Phase 2 ships only when **all eight** acceptance criteria (P2-AC1..AC8) and **all five** SLOs pass on a clean multi-region baseline (§11.4.40 full-suite retest, §11.4.108 runtime-signature on a fresh fleet deploy, §11.4.129 huge-blocker full-restart if a release-blocker surfaces). These are restated from [04_P2 §10.2/§10.3] and bound to work-items in §18. **The SLOs are themselves acceptance** — a feature is not "done" until its SLO histogram is green and alert-wired.

graph LR

```

subgraph DoD["Phase-2 Definition-of-Done (release gates)"]
  A1["P2-AC1 full transport set<br/>+ auto-ladder + per-net
memory<br/>+ regional priors thru hostile net"]
  A2["P2-AC2 DAITA toggle<br/>measurably drops
fingerprinting<br/>cost surfaced honestly"]
  A3["P2-AC3 direct P2P behind NAT<br/>seamless relay
fallback<br/>live path indicator"]
  A4["P2-AC4 multi-hop entry/exit<br/>jurisdiction-separable<br/>
privacy proven by capture"]
  A5["P2-AC5 PQ handshake default<br/>hybrid never PQ-only<br/>
zero steady-state cost"]
end

```

```

A6["P2-AC6 Win+macOS desktop<br/>parity incl. split-tunnel"]
A7["P2-AC7 GitOps PR<br/>converges fleet < 1s<br/>reviewed
effect-diff"]
A8["P2-AC8 survive regional outage<br/>reconnect < 3s,<br/>no
control-plane data loss"]
end

```

P2-AC1 — Full transport set. All transports present (plain-UDP, LWO-hardened, MASQUE/H3, Shadowsocks-wrap, UDP-over-TCP); the refined auto-ladder with per-network memory and coordinator-pushed regional priors keeps a tunnel up through a hostile-network simulation. *Evidence:* DPI-gauntlet E2E + tshark classifier confirming each disguise + per-network-memory replay [04_P2 §1, research-hysteria2].

P2-AC2 — DAITA efficacy. With DAITA on, a closed-world website-fingerprinting classifier's accuracy drops materially vs DAITA-off, *measured* in the harness; the bandwidth/latency cost is surfaced honestly in-app. *Evidence:* the fingerprinting harness report (accuracy delta) + cover-traffic overhead metric [04_P2 §2, research-daita_test].

P2-AC3 — Direct P2P + relay fallback. A direct path establishes between client and connector behind common NATs; symmetric-NAT pairs fall back to helix-relay seamlessly; the UI shows the live direct/relay path. *Evidence:* NAT-matrix E2E (pcap proving direct datagrams bypass the gateway) + relay fallback capture [04_P2 §3].

P2-AC4 — Multi-hop. A Client → Entry → Exit chain proves entry cannot observe destination and exit cannot observe client IP, with entry/exit in different regions. *Evidence:* per-hop packet capture asserting the isolation invariants [04_P2 §4].

P2-AC5 — Post-quantum handshake. PQ-PSK is on by default where both ends support it (negotiated capability), **always hybrid** (classical + PQ, never PQ-only), with zero steady-state per-packet cost. *Evidence:* interop matrix (classical↔PQ negotiation) + downgrade-safety proof + per-packet cost benchmark [04_P2 §5, research-pki_pq_nat].

P2-AC6 — Desktop apps. Windows (wireguard-nt + privileged service) and macOS (NEPacketTunnelProvider) reach Linux parity including split-tunnel. *Evidence:* per-platform UX recordings driving connect → reach → split-tunnel [04_P2 §6, §11.4.159].

P2-AC7 — GitOps. A merged policy PR converges the whole fleet in < 1 s with a CI-rendered *effect-diff* (which devices gain/lose which access) reviewed before merge. *Evidence:* PR pipeline log + helix_reconcile_seconds p99 + the effect-diff artefact [04_P2 §7].

P2-AC8 — Regional outage survival. Killing a whole region reconnects affected clients in < 3 s with zero policy/identity loss; existing direct-P2P sessions survive untouched. *Evidence:* chaos run capture (region down → reconnect timing

- post-recovery state integrity) [04_P2 §8/§10.1].

SLOs (measured, alert-on-breach) — these ARE acceptance

ID	SLO	Target	Item that wires the metric
P2-SLO1	direct-path establishment (non-symmetric NAT)	> 90% within 10 s	HVPN-P2-224
P2-SLO2	relay → direct upgrade (when possible)	< 15 s, no user-visible reconnect	HVPN-P2-225
P2-SLO3	gateway failover → client reconnected	p99 < 3 s	HVPN-P2-274
P2-SLO4	cross-region event propagation	p99 < 2 s	HVPN-P2-272
P2-SLO5	PQ session adoption (capable pairs)	> 95%	HVPN-P2-243

3. Workable-item schema (§11.4.93 DB-ready)

Every leaf in §§6–16 maps 1:1 to a row in the git-tracked SQLite single source of truth (docs/.workable_items.db, §11.4.93/.95). The Phase-1 loader (cmd/workable-items/, HVPN-P1-150) parses these leaf blocks and upserts by atm_id; Phase 2 reuses it unchanged (the schema below is the same DDL the Phase-1 WBS §3 declared, restated for self-containment). The phase discriminator column lets one DB hold both phases.

```
-- docs/.workable_items.db – Phase-2 WBS projection (same schema
as [07] §3)
CREATE TABLE IF NOT EXISTS items (
  atm_id      TEXT PRIMARY KEY,          -- 'HVPN-
P2-221'
  parent_id   TEXT REFERENCES items(atm_id), -- epic or
parent task ('HVPN-P2-E22')
  phase       TEXT NOT NULL DEFAULT 'P2' CHECK (phase IN
('P0', 'P1', 'P2', 'P3')),
  kind        TEXT NOT NULL CHECK (kind IN
('epic', 'task', 'subtask')),
  title       TEXT NOT NULL CHECK (length(title) >= 6),
  description  TEXT NOT NULL CHECK (length(description) >= 40),
-- §11.4.91/.148
  status      TEXT NOT NULL DEFAULT 'Queued'
CHECK (status IN ('Queued', 'In progress', 'Ready
for testing',
                  'In
testing', 'Reopened', 'Operator-blocked',
                  'Obsolete (→
Fixed.md)', 'Implemented (→ Fixed.md)',
                  'Completed (→ Fixed.md)', 'Fixed
(→ Fixed.md)')),
```

```

type          TEXT NOT NULL DEFAULT 'Feature' CHECK (type IN
              ('Bug','Feature','Task')),
severity      TEXT NOT NULL DEFAULT 'normal',
epic          TEXT NOT NULL,                      -- 'E22'
module        TEXT NOT NULL,                      -- 'helix-
              transport','helix-relay','coordinator',...
deps          TEXT NOT NULL DEFAULT '[]',          -- JSON array
              of atm_ids (cross-phase allowed)
deliverable   TEXT NOT NULL,
acceptance    TEXT NOT NULL,
effort_days    INTEGER NOT NULL,
test_types     TEXT NOT NULL,                     -- JSON:
              ["UNIT","INT","SEC","DDOS"] (§11.4.169)
dod_refs      TEXT NOT NULL DEFAULT '[]',          -- JSON: ["P2-
              AC3","P2-SLO1"]
source_refs    TEXT NOT NULL DEFAULT '[]',          -- JSON:
              ["04_P2 §3","11_MST §4"]
created_at     TEXT NOT NULL DEFAULT (datetime('now')),
modified_at    TEXT NOT NULL DEFAULT (datetime('now'))
);
-- per-item test diary (§11.4.149) – a PASS row REQUIRES a non-
  empty evidence path:
CREATE TABLE IF NOT EXISTS test_diary (
  id           INTEGER PRIMARY KEY AUTOINCREMENT,
  atm_id       TEXT NOT NULL REFERENCES items(atm_id),
  date_time    TEXT NOT NULL,
  tested_by    TEXT NOT NULL CHECK (tested_by IN
              ('User','Operator','AI-agent','HelixQA')),
  result       TEXT NOT NULL CHECK (result IN
              ('PASS','FAIL','SKIP')),
  test_type    TEXT NOT NULL,                      -- one of the
              §11.4.169 vocab
  evidence_path TEXT,                              -- qa-results/
              <run-id>/...
  observations TEXT NOT NULL,
  CHECK (result <> 'PASS' OR (evidence_path IS NOT NULL AND
              length(evidence_path) > 0))
);

```

Example row (one leaf) so the projection is unambiguous:

```

INSERT INTO items
  (atm_id,parent_id,phase,kind,title,description,epic,module,deps,
   deliverable,acceptance,effort_days,test_types,dod_refs,source_refs)
VALUES (
  'HVPN-P2-222','HVPN-P2-E22','P2','task',
  'UDP hole punching across NAT-type matrix',
  'Drive simultaneous WG handshake probes to every peer candidate
   so full-/restricted-/port-restricted-cone NATs open a
   direct path, latching WireGuard roaming onto the working
   endpoint.',
  'E22','helix-core','["HVPN-P2-220","HVPN-P2-221"]',
  'helix-core/src/nat/punch.rs; netns NAT-matrix harness',

```

```
'Direct datagrams captured bypassing the gateway for full/
restricted/port-restricted pairs; symmetric pairs cleanly
fall through to relay (HVPN-P2-223); no false direct
claim.',
8,['UNIT',"E2E","STRESS","SEC","CHAL"],['P2-AC3',"P2-
SL01"],['04_P2 §3.2',"SYNTHESIS §3 D6"]');
```

4. Epic map + dependency graph

graph TD

```
P1["Phase 1 MVP shipped<br/>(8 DoD + 4 SLO green)"] --> E20
P1 --> E22
P1 --> E24
P1 --> E27

E20["E20 Full transport set<br/>Shadowsocks · UoT · LW0+ ·
ladder"] --> E21
E20 --> E29
E21["E21 DAITA / maybenot<br/>traffic-analysis defense"] --> E29
E22["E22 Direct P2P + NAT traversal<br/>STUN · hole-punch ·
helix-relay"] --> E23
E22 --> E28
E22 --> E29
E23["E23 Multi-hop<br/>nested WireGuard"] --> E29
E24["E24 Post-quantum handshake<br/>ML-KEM PSK · hybrid"] -->
E29
E25["E25 Desktop apps<br/>Windows + macOS"] --> E29
E26["E26 Policy-as-code / GitOps"] --> E29
E27["E27 HA + multi-region<br/>Patroni · NATS JetStream ·
steering"] --> E22
E27 --> E26
E27 --> E28
E28["E28 Observability additions"] --> E29
E29["E29 QA · SLOs · DoD cert"] --> E30
E30["E30 docs-chain · items DB · release tag"]

classDef crit fill:#fde,stroke:#c33,stroke-width:2px;
class E22,E24,E27,E21,E29 crit;
```

Critical path (red): **E27** → **E22** → **E23** → **E29** for the networking/reach spine, with **E24 (PQ)** and **E21 (DAITA)** as parallel high-risk tracks that converge at E29. E27 (multi-region) leads because the STUN endpoints, relay fleet, and NATS event mesh that E22/E26/E28 consume are *its* outputs (a single-region P2P can be prototyped earlier, but production NAT traversal + relay fungibility depend on the multi-region edge). E20/E25/E26 fan out as disjoint-scope PWUs (§11.4.58) off the shipped MVP. Audio-style "top priority first" (§11.4.132) maps here to the **correctness-and-downgrade-safety floor**: NAT-traversal honesty (E22), DAITA *measured* efficacy (E21), PQ *never-weaker-than-classical* (E24) lead every iteration.

5. Phase-2 entry condition (Phase 1 shipped)

Phase 2 is strictly additive; it assumes the Phase-1 seams exist and are green. These items are **not** Phase-2 work but the gating precondition — they track that the reserved seams are present so no Phase-2 capability has to reshape an interface (§11.4.114 — the shipped MVP tag is the known-good baseline).

HVPN-P2-E19 — Phase-1 seam certification (entry gate) epic · module: gate

- **HVPN-P2-190 — Transport-trait seam present.** XS(1) · deps: HVPN-P1-090 · tests: UNIT, INT
 - *Desc:* Confirm the Phase-1 Transport trait + TransportPolicy.order ladder exist and accept new impls without signature change, so E20 is purely additive. *Deliverable:* seam-audit report citing helix-transport/src/lib.rs. *Acceptance:* a no-op stub transport registers + the ladder walks it with zero changes to the trait [04_P0, 04_P1 §5].
 - **HVPN-P2-191 — Peer.endpoint candidate-list seam present.** XS(1) · deps: HVPN-P1-071 · tests: UNIT
 - *Desc:* Confirm the NetworkMap.Peer.endpoint already carries a *candidate list* (not a single addr) reserved in Phase 1 for NAT traversal. *Deliverable:* protobuf audit. *Acceptance:* the field is repeated EndpointCandidate; the coordinator can emit a delta populating it [04_P2 §3.1].
 - **HVPN-P2-192 — Federated Coordinator + PKI hooks present.** XS(1) · deps: HVPN-P1-070, HVPN-P1-030 · tests: UNIT, SEC
 - *Desc:* Confirm coordinator statelessness assumptions (rebuildable cache) and PKI PSK hooks reserved in Phase 1 so HA (E27) and PQ (E24) attach without redesign. *Deliverable:* seam audit. *Acceptance:* coordinator hydrates from store+events with no local durable state; PKI exposes a PSK-injection hook [04_ARCH §10, 04_P2 §5.1].
-

6. E20 — Full transport / obfuscation set

Phase 1 proved the Transport trait with three impls (plain-UDP, basic LWO, MASQUE/H3). Phase 2 fills the matrix — **every new transport is just another impl of the same trait**, shared byte-for-byte across client, connector, and edge [04_P2 §1]. The trait the new impls satisfy (restated from [01]/[04_P0]):

```
// helix-transport/src/lib.rs – the seam (unchanged since Phase 1)
#[async_trait::async_trait]
pub trait Transport: Send + Sync {
    /// Stable, ladder-addressable id, e.g. "plain-udp", "masque-
    h3", "shadowsocks", "udp-over-tcp".
    fn id(&self) -> &'static str;
    /// Dial the peer endpoint; returns a duplex framed channel
    carrying WG datagrams.
    async fn dial(&self, ep: &Endpoint, params: &TransportParams)
    -> Result<Channel, TransportError>;
    /// Listen (edge/connector side) for inbound disguised WG
    datagrams.
```

```

    async fn accept(&self, bind: &Endpoint, params:
        &TransportParams) -> Result<Channel, TransportError>;
    /// Reliability class – informs the ladder & DAITA timing
    model.
    fn reliability(&self) -> Reliability; // Datagram { lossy } |
        Stream { hol_blocking }
}

```

HVPN-P2-E20 — Full transport / obfuscation set epic · module: helix-transport

- **HVPN-P2-200 — Shadowsocks-wrap transport.** M(6) · deps: HVPN-P2-190 · tests: UNIT, INT, E2E, SEC, BENCH, CHAL
 - *Desc:* Wrap WG datagrams in a Shadowsocks AEAD stream (looks like random/TLS-ish TCP) for China-style DPI where even QUIC is throttled, reusing shadowsocks-rust crypto primitives rather than re-implementing AEAD framing. *Deliverable:* helix-transport/src/shadowsocks.rs (sketch below); DPI-gauntlet fixture. *Acceptance:* tunnel survives udp drop + quic drop nft rules; tshark classifier sees no WG/QUIC signature; goodput recorded with the TCP-HoL caveat noted [04_P2 §1.1, research-hysteria2].

```

// helix-transport/src/shadowsocks.rs
pub struct ShadowsocksTransport {
    cipher: AeadKind,           // ChaCha20-Poly1305 | Aes256Gcm
    psk: SecretBytes,          // transport password – SEPARATE
                               // from WG keys (§04)
}

#[async_trait::async_trait]
impl Transport for ShadowsocksTransport {
    fn id(&self) -> &'static str { "shadowsocks" }
    async fn dial(&self, ep: &Endpoint, p: &TransportParams) ->
        Result<Channel, TransportError> {
        let tcp = TcpStream::connect(ep.sockaddr()).await?;
        let aead = AeadStream::client(tcp, self.cipher,
            &self.psk)?; // salt + AEAD session

        Ok(Channel::stream(LengthPrefixedWg::new(aead))) //
            2-byte len + WG datagram
    }
    async fn accept(&self, bind: &Endpoint, p: &TransportParams)
        -> Result<Channel, TransportError> { /* mirror */ }
    fn reliability(&self) -> Reliability { Reliability::Stream {
        hol_blocking: true } }
}

```

- **HVPN-P2-201 — UDP-over-TCP (UoT) transport.** S(3) · deps: HVPN-P2-190 · tests: UNIT, E2E, BENCH
 - *Desc:* Length-prefix WG datagrams over a single TCP connection as the last-resort transport when *all* UDP is blocked, accepting head-of-line-blocking purely to keep a tunnel *possible* (matches Mullvad udp2tcp). *Deliverable:* helix-transport/src/udp_over_tcp.rs. *Acceptance:* tunnel comes up under udp drop (all ports); reachability preserved; HoL penalty measured + documented; ladder ranks it last [04_P2 §1.2].

- **HVPN-P2-202 — LWO hardening (per-session keyed header obfuscation).**

M(5) · deps: HVPN-P2-190 · tests: UNIT, SEC, BENCH, CHAL

- *Desc:* Upgrade the Phase-1 XOR/padding LWO to a per-session-keyed scheme obfuscating exactly the WG header bytes DPI signatures key on (message-type/reserved fields) plus randomized padding — cheap evasion of *naive* WG fingerprinting without QUIC overhead (Mullvad LWO is the reference). *Deliverable:* helix-transport/src/lwo.rs v2. *Acceptance:* a WG-signature DPI rule that drops Phase-1 plain-WG fails to classify LWO-v2 traffic; overhead < 2% vs plain-UDP [04_P2 §1.3].

- **HVPN-P2-203 — Refined auto-escalation ladder + per-network memory.**

M(6) · deps: HVPN-P2-200, HVPN-P2-201, HVPN-P2-202 · tests: UNIT, E2E, FA, CHAL

- *Desc:* Walk TransportPolicy.order on repeated handshake failure across the full set with a per-step failure budget (N handshakes / T seconds), and **remember the working transport per-network** (SSID/gateway fingerprint) so reconnects on the same hostile network skip straight to what worked. *Deliverable:* helix-core/src/ladder.rs FSM + network_memory store. *Acceptance:* on a network that blocks UDP+QUIC the client converges to shadowsocks, persists it, and on the *next* connect to the same network skips escalation latency (captured timing delta) [04_P2 §1.4].

```
// helix-core/src/ladder.rs - escalation FSM (sketch)
pub struct Ladder { order: Vec<TransportId>, budget:
    FailureBudget, memory: NetworkMemory }
impl Ladder {
    pub async fn connect(&mut self, net: NetworkFingerprint) ->
        Result<ActiveTransport, LadderError> {
        // 1. fast-path: a remembered transport for THIS network
        // jumps the queue
        let start = self.memory.get(&net).map(|t|
            self.index_of(t)).unwrap_or(0);
        for t in &self.order[start..] {
            match self.try_handshake(t, &self.budget).await {
                Ok(active) => { self.memory.remember(net, *t);
                return Ok(active); }
                Err(_) if self.budget.exhausted() =>
                continue, // escalate
                Err(e) => return
                Err(e), // hard fail (not
                censorship)
            }
        }
        Err(LadderError::AllExhausted)
    }
}
```

- **HVPN-P2-204 — Coordinator-pushed regional priors.** S(3) · deps:

HVPN-P2-203, HVPN-P1-072 · tests: UNIT, INT, E2E

- *Desc:* Let the coordinator push a TransportPolicy.order prior keyed on the client's resolved region (e.g. "CN-resolved clients start at shadowsocks") inside the NetworkMap, so users in censored regions skip escalation latency. *Deliverable:* coordinator/policy/regional_prior.go + TransportPolicy.region_prior map field. *Acceptance:* a client whose probe IP resolves to a censored region receives

a reordered ladder in its first map and connects on the prior's head transport [04_P2 §1.4].

- **HVPN-P2-205 — Aggregate censorship-evasion telemetry.** S(2) · deps: HVPN-P2-203, HVPN-P1-080 · tests: UNIT, SEC, CHAL
 - *Desc:* Record **only** "transport X succeeded after N escalations in region R" (aggregate, no per-user data) feeding the Censorship-Evasion Success dashboard, with a CI schema-lint asserting no per-connection row appears (§11.4 no-logging). *Deliverable:* telemetry/evasion.go + Grafana panel. *Acceptance:* dashboard renders transport-mix by region; the §11.4 schema-lint mutation (a per-user evasion table) FAILs the build [04_ARCH §9, 04_P2 §1.4].
 - **HVPN-P2-206 — Hysteria2/Salamander interop evaluation (decision D1/D6).** S(3) · deps: HVPN-P2-200 · tests: BENCH, E2E
 - *Desc:* Benchmark a Hysteria2+Salamander obfuscated-QUIC leg vs MASQUE/H3 on the user[?]gateway hop, to decide whether the *asymmetric per-leg* topology [11_MST] (Hysteria2 user[?]gateway, WG gateway[?]networks) is worth adopting over the single-protocol MASQUE end-to-end stance [04_ARCH §3.3]. *Deliverable:* §20 decision-log row + A/B CSV (goodput under loss, brutal-congestion behaviour, DPI evasion parity). *Acceptance:* a recorded decision with evidence; **recommendation: keep MASQUE/H3 primary** (single Rust impl, true Mullvad-parity, no second protocol stack) and treat Hysteria2 as an *optional* ladder rung only if the benchmark shows a material reach/goodput win in hostile networks [SYNTHESIS §3 D1/D6, research-hysteria2, research-masque].
-

7. E21 — DAITA traffic-analysis defense (maybenot)

Even fully encrypted, packet **size/timing/frequency** patterns fingerprint which site a user visits. DAITA defeats this. **Do not roll your own** — adopt the **maybenot** framework (the engine behind Mullvad's DAITA): a lightweight state-machine engine for padding + cover-traffic "machines" [04_P2 §2, research-daita_test]. DAITA shaping is a layer **above WireGuard, below the transport**, orthogonal to *which* transport is in use:

TUN → WG encrypt → [DAITA shaping: pad + inject cover] → Transport → wire

HVPN-P2-E21 — DAITA / maybenot epic · module: helix-core

- **HVPN-P2-210 — maybenot engine integration.** M(6) · deps: HVPN-P2-203 · tests: UNIT, INT, BENCH
 - *Desc:* Embed the maybenot::Framework as a shaping stage over the WG-datagram stream: per-outgoing-datagram on_packet may pad and/or schedule cover packets, and a timer-driven tick emits scheduled padding/cover — machines are **config (data), not code**. *Deliverable:* helix-core/src/daita.rs (sketch below). *Acceptance:* with a known padding machine loaded, outgoing datagrams are padded to uniform sizes and cover packets emit on schedule (capture asserts size histogram collapses to the target sizes) [04_P2 §2.3].

```
// helix-core/src/daita.rs — shaping stage (sketch)
pub struct Daita { framework: maybenot::Framework /* machines =
    NetworkMap data */ }
impl Daita {
    fn on_packet(&mut self, dg: &mut Bytes, now: Instant) ->
        Vec<ScheduledAction> { /* pad + schedule cover */ }
    fn tick(&mut self, now: Instant) -> Vec<Bytes> { /* emit due
        cover/padding packets */ }
}
```

- **HVPN-P2-211 — Machines-as-data distribution via NetworkMap.** S(3) ·
 deps: HVPN-P2-210, HVPN-P1-071 · tests: UNIT, INT, E2E
 - *Desc:* Add a daita field to NetworkMap carrying signed maybenot machine definitions so defenses are tuned/updated server-side without shipping a client build, gated by the architecture parity matrix. *Deliverable:* proto/networkmap.proto DaitaConfig + coordinator/daita.go. *Acceptance:* a coordinator-pushed machine update changes client shaping behaviour live (no rebuild); machine blobs are signature-verified before load (tampered blob rejected, captured) [04_P2 §2.3, §04].
 - **HVPN-P2-212 — DAITA opt-in toggle + honest cost surface.** S(3) ·
 deps: HVPN-P2-210, HVPN-P1-110 · tests: UI, UX, REC
 - *Desc:* Surface DAITA as a "maximum privacy" opt-in (off by default, Mullvad's stance) in the Access app with an honest cost note (bandwidth/latency), reflecting the live cover_ratio. *Deliverable:* helix-ui DaitaToggle widget + Riverpod binding to core status. *Acceptance:* UX recording shows toggling DAITA on/off, the cost note, and the live overhead readout (§11.4.159 vision-verified) [04_P2 §2.3].
 - **HVPN-P2-213 — Closed-world fingerprinting efficacy harness (the privacy claim).** L(8) ·
 deps: HVPN-P2-210 · tests: FA, SEC, CHAL, BENCH
 - *Desc:* Build the *measure-don't-assert* harness: a closed-world website-fingerprinting classifier trained on captured traces, run DAITA-off vs DAITA-on, asserting classifier accuracy drops *materially* — with the analyzer itself self-validated (golden-good machine reduces accuracy; golden-bad "no-op machine" does NOT, §11.4.107(10)). *Deliverable:* tests/daita/wf_harness/ + accuracy report. *Acceptance:* DAITA-on accuracy < DAITA-off accuracy by the documented margin on the project's own fixtures; the golden-bad no-op machine FAILs the analyzer (proving the harness can't bluff); cover-traffic overhead recorded [04_P2 §2.3/§10.1, research-daita_test, §11.4.123].
-

8. E22 — Direct P2P + NAT traversal + helix-relay

Phase 1 relayed **all** peer traffic through the gateway. Phase 2 makes the gateway a **coordinator + relay-of-last-resort**, with traffic going **directly** between client and connector wherever NAT allows (the Tailscale/WireGuard roaming model) — lower latency, far less gateway bandwidth, better scaling [04_P2 §3].

HVPN-P2-E22 — Direct P2P + NAT traversal + relay epic · module: helix-core, helix-relay, coordinator

- **HVPN-P2-220 — STUN-like endpoint discovery.** M(5) · deps: HVPN-P2-191, HVPN-P2-271 · tests: UNIT, INT, E2E, DDOS
 - *Desc:* Each node learns its candidates — all local interface addresses (host candidates) + a server-reflexive candidate discovered by probing the gateway's STUN-like endpoint (which echoes the observed src ip:port, revealing the NAT mapping) — reported to the coordinator via ReportStatus. *Deliverable:* helix-core/src/nat/discover.rs + helix-edge/src/stun.rs. *Acceptance:* a node behind NAT reports a correct server-reflexive ip:port; the STUN endpoint refuses amplification (response ≤ request size, rate-limited — DDoS test) [04_P2 §3.1, §04].
- **HVPN-P2-221 — Candidate distribution via WatchNetworkMap (no new signal service).** S(3) · deps: HVPN-P2-220 · tests: UNIT, INT, E2E
 - *Desc:* Reuse the existing WatchNetworkMap stream **as the signaling channel**: when two authorized peers come online the coordinator pushes each the other's candidate list as a Peer.endpoint delta (NetBird uses a dedicated signal service; HelixVPN folds it into the coordinator). *Deliverable:* coordinator/signaling.go (delta builder). *Acceptance:* both peers receive each other's candidates within the SLO1 budget; candidates are policy-filtered (need-to-know — a peer not authorized never appears) [04_P2 §3.3, §04].
- **HVPN-P2-222 — UDP hole punching across the NAT-type matrix.** L(8) · deps: HVPN-P2-220, HVPN-P2-221 · tests: UNIT, E2E, STRESS, SEC, CHAL
 - *Desc:* Both peers simultaneously send WG handshake probes to all of the peer's candidates; for full-/restricted-/port-restricted-cone NATs a path opens and WireGuard's roaming latches onto whatever source a valid handshake arrives from; symmetric-on-both-ends fails over to relay. *Deliverable:* helix-core/src/nat/punch.rs + netns NAT-matrix harness. *Acceptance:* direct datagrams **captured bypassing the gateway** for full/restricted/port-restricted pairs; symmetric pairs cleanly fall through to relay; **no false "direct" claim** (a path is "direct" only when a non-gateway src is observed at the peer, §11.4.107) [04_P2 §3.2].

```
// helix-core/src/nat/punch.rs - simultaneous-open (sketch)
pub async fn punch(peer: &PeerCandidates, wg: &WgHandle) ->
    PathOutcome {
    // fire WG handshake initiations at EVERY candidate
    // concurrently
    let probes = peer.candidates.iter().map(|c|
        wg.send_handshake_to(c));
    join_all(probes).await;
    // WG roams to whatever valid response source arrives first
    match wg.await_first_valid_response(HOLE_PUNCH_TIMEOUT).await
    {
        Some(src) if !src.is_gateway() =>
            PathOutcome::Direct(src), // <-- proven by non-gateway
            src
        =>
            PathOutcome::RelayFallback, // symmetric / hostile FW
    }
}
```

}
}

- **HVPN-P2-223 — helix-relay (DERP-style encrypted relay fallback).**
L(8) · deps: HVPN-P2-222, HVPN-P2-273 · tests: UNIT, INT, E2E, SEC, STRESS, BENCH
 - *Desc:* When direct fails (symmetric NAT / hostile firewall), traffic relays through the edge in helix-relay mode over an encrypted relay keyed by destination public key — the relay sees only encrypted WG datagrams **below the WG crypto boundary** (it cannot read traffic), preserving no-logging even on the relay path; relays are fungible across tenants/regions. *Deliverable:* helix-edge/src/relay.rs (mode of the edge). *Acceptance:* symmetric-NAT pair communicates via relay; a memory/pcap inspection of the relay proves it holds only ciphertext keyed by pubkey (no plaintext, no per-flow durable record); any region's relay forwards for any tenant [04_P2 §3.4, §04].
- **HVPN-P2-224 — Path selection: start-relay, upgrade-to-direct.** M(6) · deps: HVPN-P2-222, HVPN-P2-223 · tests: UNIT, E2E, PERF, FA · DoD: P2-SLO1
 - *Desc:* Start on relay for instant connectivity, attempt direct in the background, and upgrade seamlessly when a direct path is confirmed (latency drop, no user-visible reconnect); report current path (direct/relay) in status. *Deliverable:* helix-core/src/path.rs selector + StatusReport.path. *Acceptance:* **P2-SLO1** — direct-path establishment for non-symmetric NAT > **90% within 10 s**, measured histogram; relay path is the instant fallback with no connectivity gap [04_P2 §3.5/§10.3].
- **HVPN-P2-225 — Continuous direct-path health-check + downgrade.** S(3) · deps: HVPN-P2-224 · tests: UNIT, E2E, PERF, CHAOS · DoD: P2-SLO2
 - *Desc:* Continuously health-check the direct path and downgrade to relay if it degrades, surfacing the current path to the UI; the relay→direct upgrade and direct→relay downgrade are both seamless (no user-visible reconnect). *Deliverable:* helix-core/src/path.rs health loop + UI PathIndicator. *Acceptance:* **P2-SLO2** — relay→direct upgrade < **15 s** with no reconnect; injected direct-path degradation downgrades to relay within the health-check window, captured [04_P2 §3.5/§10.3].
- **HVPN-P2-226 — Path indicator UI + UX.** S(2) · deps: HVPN-P2-224, HVPN-P1-110 · tests: UI, UX, REC
 - *Desc:* Show the live direct/relay path (and NAT-traversal state) in the Access app via a PathIndicator component bound to the core status stream. *Deliverable:* helix-ui PathIndicator + golden tests. *Acceptance:* UX recording shows the indicator flipping relay→direct on upgrade and direct→relay on injected degradation (§11.4.159 vision-verified) [04_P2 §3.5].

9. E23 — Multi-hop (nested WireGuard)

Generalizes §3.5 of the architecture: a client routes Client → Entry → Exit → {internet | connector} so no single node sees both who you are and where you're going [04_P2 §4]. Mechanism — **nested WireGuard**, no new wire protocol:

Client holds TWO WG sessions:

outer: Client ↔ Entry (Entry sees client IP, not destination)

inner: Client ↔ Exit (Exit sees destination, not client IP – only sees Entry)
packets: WG_outer(WG_inner(payload)) → Entry decapsulates
outer → forwards to Exit
transport/obfuscation applies to the OUTER session; inner is plain WG inside it.

HVPN-P2-E23 — Multi-hop (nested WireGuard) epic · module: helix-core, coordinator

- **HVPN-P2-230 — Hop-chain computation in the coordinator.**

M(5) · deps: HVPN-P2-191, HVPN-P1-072 · tests: UNIT, INT, SEC

- *Desc*: Compute the hop chain (respecting exitNodes policy + user selection), assign per-hop keys, and push a multi-hop NetworkMap with a hops list; entry/exit may be gateways in different regions/jurisdictions (the privacy point). *Deliverable*: proto/networkmap.proto Hop/hops + coordinator/hops.go. *Acceptance*: a user choosing Entry=EU, Exit=US receives a two-hop map with distinct per-hop keys; policy exitNodes is enforced (an unauthorized exit is never offered) [04_P2 §4.2].

- **HVPN-P2-231 — Nested-session construction in helix-core.** M(6) · deps: HVPN-P2-230 · tests: UNIT, E2E, BENCH

- *Desc*: Build the nested WG sessions from the hops map (outer Client[?]Entry obfuscated, inner Client[?]Exit plain-WG inside), with the transport/DAITA stack applied to the outer session only. *Deliverable*: helix-core/src/multihop.rs. *Acceptance*: end-to-end reachability through Entry→Exit; goodput/latency overhead of the second hop measured + documented [04_P2 §4.1].

- **HVPN-P2-232 — Hop-isolation capture assertions (the privacy proof).**

M(5) · deps: HVPN-P2-231 · tests: SEC, E2E, CHAL

- *Desc*: Packet-capture at each hop proving Entry observes client IP but not destination, and Exit observes destination but not client IP (it only sees Entry) — the actual multi-hop privacy claim, measured not asserted. *Deliverable*: tests/multihop/isolation/ capture harness. *Acceptance*: **P2-AC4** — at Entry, destination is absent from the cleartext it can see; at Exit, the client IP is absent (only Entry's appears); both proven by pcap, with a mutation (collapse to single hop) FAILing the assertion [04_P2 §4, §10.1].

- **HVPN-P2-233 — Multi-hop picker UI.** S(3) · deps: HVPN-P2-230, HVPN-P1-110 · tests: UI, UX, REC

- *Desc*: Add an entry/exit picker (region/jurisdiction labelled) to the Access app, bound to the policy-permitted hop set. *Deliverable*: helix-ui MultiHopPicker. *Acceptance*: UX recording shows selecting Entry/Exit and the resulting two-hop connection with the path/jurisdiction labels (§11.4.159) [04_P2 §4].

10. E24 — Post-quantum handshake (ML-KEM PSK)

WireGuard's Curve25519 handshake is vulnerable to **harvest-now-decrypt-later**. Phase 2 closes this the way Mullvad does — **without forking WireGuard's crypto** — by establishing a post-quantum-derived **pre-shared key (PSK)** over the already-authenticated control channel and mixing it into the WG handshake [04_P2 §5, research-pki_pq_nat]:

on session setup, over the (already-authenticated) Coordinator RPC:

client → gateway: PQ-KEM public key (ML-KEM / Kyber, FIPS 203)

gateway → client: KEM ciphertext

both derive: shared secret → HKDF → WG PSK

WG handshake then proceeds with that PSK mixed in. rotate on each rekey.

HVPN-P2-E24 — Post-quantum handshake epic · module: pki, helix-core

- **HVPN-P2-240 — ML-KEM PSK exchange over the control channel.** L(8)

- deps: HVPN-P2-192 · tests: UNIT,INT,SEC,BENCH

- *Desc*: Implement the ML-KEM (FIPS 203) KEM exchange riding the existing authenticated Coordinator RPC (no new public listener, no unauthenticated PQ endpoint to attack), deriving the WG PSK via HKDF and rotating it on each rekey interval. *Deliverable*: pki/pq/mlkem.rs + helix-core/src/pq.rs (sketch below). *Acceptance*: two PQ-capable peers establish a WG session whose PSK is ML-KEM-derived; PSK rotates on rekey (captured); the exchange happens at setup/rekey only [04_P2 §5.1, research-pki_pq_nat].

```
// helix-core/src/pq.rs – PQ-PSK negotiation (sketch)
pub enum KemSuite { MlKem768, MlKem768PlusMcEliece /* hybrid-conservative hedge */ }
pub trait PqKem { fn keypair(&self) -> (PqPublic, PqSecret);
                  fn encapsulate(&self, pk: &PqPublic) -> (Ciphertext, SharedSecret);
                  fn decapsulate(&self, ct: &Ciphertext, sk: &PqSecret) -> SharedSecret; }
pub fn derive_wg_psk(classical: &SharedSecret, pq: &SharedSecret)
    -> WgPsk {
    // HYBRID: PSK = HKDF(classical || pq) – a flaw in EITHER cannot weaken below the other (§AC5)
    Hkdf::<Sha256>::new(None, &[classical.as_bytes(),
    pq.as_bytes()].concat()).expand_psk()
}
```

- **HVPN-P2-241 — Hybrid combiner (never PQ-only) + capability**

- negotiation.**M(5) · deps: HVPN-P2-240 · tests: UNIT,SEC,E2E

- *Desc*: Always combine the PQ secret with the classical exchange (defense-in-depth: an attacker must break *both*; a flaw in the younger PQ primitive can't *weaken* you below classical), and negotiate the capability so classical-only ☐ PQ-capable pairs interoperate. *Deliverable*: helix-core/

src/pq.rs combiner + proto capability flags. *Acceptance*: PQ session never derives the PSK from PQ alone (unit + mutation: PQ-only combine → FAIL); a PQ-capable client to a classical-only gateway negotiates down to classical without error [04_P2 §5.1].

- **HVPN-P2-242 — Downgrade-safety guard.**

M(5) · deps: HVPN-P2-241 · tests: SEC, E2E, CHAL

- *Desc*: Guarantee a PQ failure **never silently drops to weaker-than-classical** — a failed PQ exchange either re-establishes classical (still today's security) or fails closed, never a degraded path; a downgrade is logged as a control event (not traffic). *Deliverable*: helix-core/src/pq.rs downgrade FSM + audit hook. *Acceptance*: injecting a PQ-exchange failure yields either a clean classical session or a fail-closed, **never** a session weaker than classical (captured); an MITM strip-PQ attempt is detected/logged [04_P2 §5.1/§10.1, §04].

- **HVPN-P2-243 — PQ default-on where supported + adoption metric.** S(3)

· deps: HVPN-P2-241, HVPN-P2-280 · tests: UNIT, PERF, REC · DoD: P2-SLO5

- *Desc*: Make PQ on by default for new tunnels where both ends advertise support (negotiated), with zero steady-state per-packet cost (KEM only at setup/rekey), and surface a "Quantum-resistant" chip; wire the adoption metric. *Deliverable*: helix-core default + helix-ui PqChip + helix_pq_sessions_ratio metric. *Acceptance*: **P2-SLO5** — PQ adoption among capable pairs > **95%** (dashboard) + per-packet benchmark shows ~0 steady-state delta vs classical [04_P2 §5.2/§10.3].

- **HVPN-P2-244 — Rosenpass alternative evaluation (decision).** S(3) ·

deps: HVPN-P2-240 · tests: BENCH, SEC

- *Desc*: Evaluate the audited external **Rosenpass** PQ key-exchange daemon (feeds WG a PSK) vs the in-house ML-KEM exchange, on audit pedigree, integration cost, and operational surface. *Deliverable*: §20 decision-log row + comparison. *Acceptance*: recorded decision; **recommendation: ship the in-house ML-KEM PSK exchange** (rides the existing authenticated channel, one fewer daemon, pluggable in pki) while keeping a Rosenpass adapter as a KemSuite variant for operators who prefer an externally-audited protocol [04_P2 §5.1, SYNTHESIS §4].

11. E25 — Desktop apps (Windows, macOS)

The Flutter UI + Rust core already build for desktop (proven on Linux in Phase 0/1); Phase 2 adds the two remaining first-class desktop tunnel shims — everything above the shim is the **same helix-ui + helix-core** [04_P2 §6].

Platform	Tunnel mechanism	Shim notes
Windows	wireguard-nt / wintun	a small Windows service runs the privileged tunnel hosting helix-core; the Flutter app talks to it over a named-pipe IPC; code-sign driver + service.
macOS	NEPacketTunnelProvider	same Network Extension model as iOS but desktop-class memory (the iOS ceiling does not bite); notarize + sign.

HVPN-P2-E25 — Desktop apps epic · module: shims/windows, shims/apple, helix-ui

- **HVPN-P2-250 — Windows privileged service + wireguard-nt shim.**
L(9) · deps: HVPN-P2-190 · tests: INT, E2E, SEC, REC
 - *Desc:* Build the Windows service that hosts helix-core over wireguard-nt/wintun, exposing a named-pipe IPC the unprivileged Flutter app drives (connect/disconnect/status), with a signed driver + signed service.
Deliverable: shims/windows/ service + helix_core_ffi host + named-pipe protocol. *Acceptance:* connect → reach an authorized LAN host from Windows; the app is unprivileged, the service holds the tunnel; driver + service Authenticode-signed (verified) [04_P2 §6].
 - **HVPN-P2-251 — macOS NEPacketTunnelProvider shim.** M(6) · deps: HVPN-P2-190 · tests: INT, E2E, SEC, REC
 - *Desc:* Build the macOS Network Extension hosting helix-core (desktop memory budget), notarized + signed, driving the same UI/core as every other platform. *Deliverable:* shims/apple/macos/ NE target. *Acceptance:* connect → reach from macOS; notarization + signature verified; UX recording (§11.4.159) [04_P2 §6].
 - **HVPN-P2-252 — Desktop split-tunnel (per-app routing).** L(8) · deps: HVPN-P2-250, HVPN-P2-251 · tests: UNIT, E2E, SEC, UX, REC
 - *Desc:* Implement per-app routing — the main net-new platform surface — via Windows WFP filters and macOS app-bound rules, with the include/exclude app set driven from the shared UI. *Deliverable:* shims/windows/wfp.rs + shims/apple/macos/split.swift + helix-ui split-tunnel screen. *Acceptance:* a designated app routes outside the tunnel while the rest stays inside (pcap proves the split per-app on both OSes); UX recording (§11.4.159) [04_P2 §6].
 - **HVPN-P2-253 — Desktop kill-switch + DNS-leak parity.** S(3) · deps: HVPN-P2-250, HVPN-P2-251 · tests: SEC, E2E, REC
 - *Desc:* Bring the Phase-1 core-state-machine kill-switch + DNS-leak protection to Windows (WFP) and macOS (NE rules) so desktop reaches Linux parity. *Deliverable:* platform firewall drivers wired to core state. *Acceptance:* on forced tunnel drop, host-side pcap shows zero plaintext egress + zero DNS leak on both OSes (§11.4.107 liveness) [04_P2 §6, §04].
-

12. E26 — Policy-as-code & GitOps

Phase 1 applied policy via Console/CLI. Phase 2 makes **Git the source of truth** [04_P2 §7]. The per-tenant policy repo + CI pipeline:

```
helixvpn-policy/
├─ policy.jsonc          # the ACL document (Phase 1 §7.1
format)
├─ networks.jsonc        # host/CIDR ↔ connector mappings
└─ ci/apply.yml          # validate → dry-run compile (effect-
diff) → apply on merge
```

§11.4.156 note: the consuming HelixVPN repos run **no active server-side CI** — the GitOps "pipeline" is implemented as a **local-runner** / **webhook-driven helixvpncctl flow** (the apply runner is the tenant's own

runner invoking `helixvpncctl`, not a GitHub Actions/GitLab pipeline on an owned repo). The `ci/apply.yml` is illustrative of the *steps*, executed by the tenant's chosen runner.

PR opened → validate schema + ``helixvpncctl policy compile --dry-run``

→ post the **diff of effects** (which devices gain/lose which access)

PR merged → ``helixvpncctl policy apply`` (authenticated via tenant API token)

→ control plane persists new version, emits `policy.compiled`

→ coordinator pushes deltas (Phase-1 flow), < 1s convergence

HVPN-P2-E26 — Policy-as-code & GitOps epic · module: `helixvpncctl`, `policy`

- **HVPN-P2-260 — `helixvpncctl policy compile --dry-run effect-diff`.**

M(6) · deps: HVPN-P1-062 · tests: UNIT,INT,SEC,CHAL

- *Desc*: Render the **compiled effect delta** (not just the text diff) for a proposed ACL change — which devices gain/lose which access — so reviewers approve the *consequence*, not the syntax (a one-line ACL change can silently grant a contractor camera-network access). *Deliverable*: `helixvpncctl/policy/effectdiff.go`. *Acceptance*: a PR adding one ACL line produces a human-readable effect-diff naming the exact devices/targets affected; a mutation hiding a granted edge FAILs the diff completeness check [04_P2 §7.2].

- **HVPN-P2-261 — GitOps apply runner + tenant-token auth.**

M(5) · deps: HVPN-P2-260,HVPN-P1-100 · tests: INT,E2E,SEC,FA · DoD: P2-AC7

- *Desc*: On merge, the tenant's runner authenticates with a tenant API token and runs `helixvpncctl policy apply`, the control plane persists the new version + emits `policy.compiled`, the coordinator pushes deltas, converging the fleet < 1 s; WG keys never live in the repo (only declarative intent); sensitive references encrypted with `sops/age`. *Deliverable*: `helixvpncctl policy apply` + the runner flow. *Acceptance*: **P2-AC7** — a merged PR converges the fleet < 1 s (`helix_reconcile_seconds` p99) with the effect-diff reviewed; the repo contains zero secrets (secret-leak audit green, §11.4.10) [04_P2 §7.1].

- **HVPN-P2-262 — GitOps round-trip safety (drift detect + rollback).**

S(3) · deps: HVPN-P2-261 · tests: INT,CHAOS,SEC

- *Desc*: Detect drift between Git intent and applied state and support a clean rollback (revert PR → re-apply previous version), with the apply being idempotent. *Deliverable*: `helixvpncctl policy diff/rollback`. *Acceptance*: an out-of-band manual change is flagged as drift; a revert PR restores the prior compiled state byte-identically (captured) [04_P2 §7].

13. E27 — HA + multi-region fleet

Phase 1 was a single VPS. Phase 2 is a **fleet**: HA control plane, gateways in multiple regions, automatic failover — *same images, only topology changes* [04_P2 §8]. The full reference topology is in §17.

HVPN-P2-E27 — HA + multi-region epic · module: coordinator, store, events, deploy

- **HVPN-P2-270 — Stateless coordinator fleet + stream re-pin.**

L(9) · deps: HVPN-P2-192 · tests: UNIT, INT, CHAOS, SCALE

- *Desc*: Run N stateless coordinators per region behind a load balancer (the topology graph is a rebuildable cache hydrated from Postgres + NATS); any coordinator can serve an agent's WatchNetworkMap stream and an agent re-pins to another on failure, resuming via known_version. *Deliverable*: coordinator/ statelessness + WatchNetworkMap resume-from-version. *Acceptance*: killing a coordinator mid-stream re-pins the agent to a sibling and resumes from known_version with zero missed deltas (captured); coordinator holds no local durable state (§11.4.108 clean-restart) [04_P2 §8.2].

- **HVPN-P2-271 — Postgres HA (Patroni) + regional read replicas.** L(8) ·

deps: HVPN-P1-020, HVPN-P2-270 · tests: INT, CHAOS, SCALE, SEC

- *Desc*: Deploy a single logical primary with synchronous replicas + automated failover (Patroni) and regional read replicas serving coordinator hydration; control-plane writes are low-volume (enroll/policy/advertise) so a single primary scales. *Deliverable*: deploy/patroni/ quadlets + replica wiring. *Acceptance*: killing the PG primary fails over automatically with no committed-write loss; regional coordinators hydrate from local read replicas; RLS/no-log invariants hold on replicas (§11.4.85) [04_P2 §8.2, §02].

- **HVPN-P2-272 — NATS JetStream cross-region event mesh (Redis swap).**

L(8) · deps: HVPN-P2-270 · tests: INT, E2E, CHAOS, SCALE, PERF ·

DoD: P2-SLO4

- *Desc*: Replace single-Redis with NATS JetStream as the cross-region event mesh — durable subjects mirror the Phase-1 stream taxonomy **1:1** (transport-only swap, as Phase 1 promised); events published in any region propagate to all coordinators. *Deliverable*: events/nats.go adapter (same EventBus interface as the Redis adapter, D3). *Acceptance*: **P2-SLO4** — cross-region event propagation **p99 < 2 s**; the taxonomy is identical to Phase 1 (no event renamed); a region partition + heal shows no event loss (durable subjects replay) [04_P2 §8.2/§10.3, SYNTHESIS §3 D3].

```
// events/bus.go – the interface is UNCHANGED from Phase 1; only
                        the impl swaps (D3)
```

```
type EventBus interface {
```

```
    Publish(ctx context.Context, subject string, ev Event) error
```

```
    Subscribe(ctx context.Context, subject string, durable
        string) (<-chan Event, error)
```

```

}
// Phase 1: RedisStreamsBus{} Phase 2: NatsJetStreamBus{} //
    same subjects, same taxonomy

```

- **HVPN-P2-273 — Edge/relay fleet + client steering (Anycast/GeoDNS + RTT list).** M(6) · deps: HVPN-P2-272 · tests: INT, E2E, PERF, DDoS
 - *Desc:* Deploy the edge/relay fleet across regions with Anycast (single IP routed to nearest edge — ideal for UDP/QUIC) and a GeoDNS fallback, and have clients receive a **candidate gateway list** in the NetworkMap to pick by measured RTT; relays are fungible (encrypted-by-pubkey). *Deliverable:* deploy/anycast/ + NetworkMap.gateways candidate list. *Acceptance:* a client picks the nearest edge by measured RTT; Anycast routes UDP/QUIC to the closest PoP; STUN/edge handshake flood is rate-limited (DDoS) [04_P2 §8.3, §04].
 - **HVPN-P2-274 — Regional failover flow + floating IPs.** M(6) · deps: HVPN-P2-273 · tests: E2E, CHAOS, PERF · DoD: P2-AC8, P2-SLO3
 - *Desc:* On a health probe detecting an edge unreachable, emit gateway.failover{from,to}, push a NetworkMap delta with the new endpoint, and have clients re-handshake (transport + PSK re-established) — existing **direct P2P sessions are unaffected** (they don't depend on the gateway); floating IPs give fast in-region failover. *Deliverable:* deploy/failover/ health prober + gateway.failover event + delta. *Acceptance:* **P2-SLO3** — gateway failover → client reconnected **p99 < 3 s**; **P2-AC8** — killing a whole region reconnects affected clients < 3 s with zero policy/identity loss and surviving direct-P2P sessions (chaos capture) [04_P2 §8.4/§10.3].
-

14. E28 — Observability additions

Extend the Phase-1 / architecture §9 stack for the new surface — **counts and health only, never destinations/flows** (the privacy invariant holds everywhere) [04_P2 §9].

HVPN-P2-E28 — Observability additions epic · module: telemetry, deploy

- **HVPN-P2-280 — Reach/evasion KPI dashboards.** M(5) · deps: HVPN-P2-205, HVPN-P2-224, HVPN-P2-243 · tests: INT, SEC, CHAL
 - *Desc:* Add transport-mix by region + escalation depth (censorship pressure), direct-vs-relay ratio (the scaling KPI), hole-punch success by NAT-type bucket, PQ adoption %, and DAITA overhead % — all aggregate, with the §11.4 schema-lint asserting no per-connection/destination row backs any panel. *Deliverable:* deploy/grafana/phase2/ Grafana-as-code + the metric exporters. *Acceptance:* dashboards render every KPI from aggregate counters; a mutation introducing a per-flow source table FAILs the no-log lint (§11.4 / §02 §2.4) [04_P2 §9].
- **HVPN-P2-281 — Multi-region health (NATS lag, PG replication, convergence p99).** S(3) · deps: HVPN-P2-272, HVPN-P2-271 · tests: INT, PERF, SCALE
 - *Desc:* Add per-region coordinator health, NATS consumer lag, Postgres replication lag, and cross-region map-convergence p99 panels + alerts. *Deliverable:* deploy/grafana/phase2/multiregion.json + alert rules.

Acceptance: the convergence-p99 panel reflects P2-SLO4 and alerts on breach; replication-lag/consumer-lag alerts fire under an injected partition (captured) [04_P2 §9].

15. E29 — QA · SLOs · Phase-2 DoD certification

The integration epic that proves the §2 release gates on a clean multi-region baseline (§11.4.40 full-suite retest, §11.4.118 enumerated discovery coverage, §11.4.135 standing regression guards). Each surface below is a *measure-don't-assert* gate; the privacy/efficacy claims (DAITA, multi-hop isolation, PQ) carry self-validated analyzers (§11.4.107(10)).

HVPN-P2-E29 — QA · SLOs · DoD certification epic · module: qa

- **HVPN-P2-290 — NAT-traversal matrix harness.** L(8) · deps: HVPN-P2-222, HVPN-P2-223 · tests: E2E, STRESS, FA, CHAL
 - *Desc:* Automate tests across NAT-type combinations (full-cone, restricted, port-restricted, symmetric) using netns + simulated NATs, asserting direct-path success where theoretically possible and clean relay fallback where not. *Deliverable:* tests/nat/matrix/ + the challenges/helix_qa bank entry. *Acceptance:* every theoretically-direct pair achieves direct (pcap bypass proof); every symmetric pair relays cleanly; no false direct claim; feeds P2-SLO1 [04_P2 §10.1].
- **HVPN-P2-291 — Obfuscation conformance gauntlet.** M(6) · deps: HVPN-P2-200, HVPN-P2-201, HVPN-P2-202 · tests: E2E, SEC, CHAL
 - *Desc:* Pass each transport through a DPI-simulation gauntlet (block UDP, block QUIC, block non-TLS) with a tshark classifier confirming the intended disguise. *Deliverable:* tests/dpi/gauntlet/. *Acceptance:* **P2-AC1** — each transport survives its target block and the classifier confirms the disguise (no WG/QUIC signature where the disguise demands none) [04_P2 §10.1].
- **HVPN-P2-292 — DAITA efficacy certification.** M(5) · deps: HVPN-P2-213 · tests: FA, SEC, CHAL
 - *Desc:* Certify the closed-world fingerprinting accuracy drop (the actual privacy claim) with the self-validated analyzer wired into the meta-test. *Deliverable:* certification report under qa-results/. *Acceptance:* **P2-AC2** — DAITA-on classifier accuracy drops materially vs off, golden-bad no-op machine FAILs the analyzer, cost surfaced [04_P2 §10.1, research-daita_test].
- **HVPN-P2-293 — PQ interop + downgrade-safety certification.** M(5) · deps: HVPN-P2-242, HVPN-P2-243 · tests: SEC, E2E, CHAL
 - *Desc:* Certify classical-only [?] PQ-capable negotiation and downgrade safety (a PQ failure must never silently drop to weaker-than-classical). *Deliverable:* interop matrix report. *Acceptance:* **P2-AC5** — all capability pairs interoperate; every injected PQ failure yields classical-or-fail-closed (never weaker), captured [04_P2 §10.1, research-pki_pq_nat].
- **HVPN-P2-294 — Multi-hop isolation certification.** S(3) · deps: HVPN-P2-232 · tests: SEC, E2E, CHAL
 - *Desc:* Certify entry cannot observe destination and exit cannot observe client IP via packet-capture assertions at each hop. *Deliverable:* isolation certification report. *Acceptance:* **P2-AC4** — both isolation invariants proven by pcap; single-hop collapse mutation FAILs [04_P2 §10.1].

- **HVPN-P2-295 — HA / chaos certification (kill coordinator, PG primary, region).** L(8) · deps: HVPN-P2-270, HVPN-P2-271, HVPN-P2-272, HVPN-P2-274 · tests: CHAOS, SCALE, PERF, FA · DoD: P2-AC8, P2-SLO3, P2-SLO4
 - *Desc:* Kill a coordinator, a Postgres primary, and a whole region; assert stream re-pin, failover timings, NATS consumer-lag recovery, and zero policy/identity loss. *Deliverable:* tests/ha/chaos/ + certification report. *Acceptance:* **P2-AC8 + P2-SLO3** (failover p99 < 3 s) + **P2-SLO4** (cross-region p99 < 2 s) all green; no control-plane data loss; direct-P2P sessions survive (captured) [04_P2 §10.1].
 - **HVPN-P2-296 — DDoS resilience certification (STUN + edge handshake flood).** M(5) · deps: HVPN-P2-220, HVPN-P2-273 · tests: DDOS, SEC, STRESS
 - *Desc:* Certify the now-public STUN endpoints + edge handshake surface refuse amplification and survive a flood with rate-limiting (the Phase-1 NOT_APPLICABLE re-armed, §11.4.6). *Deliverable:* tests/ddos/ + report. *Acceptance:* STUN response ≤ request (no amplification); edge sheds a handshake flood without dropping legitimate tunnels below SLO; captured [04_P2 §9, §04].
 - **HVPN-P2-297 — Phase-2 DoD certification report + discovery sweep.** M(6) · deps: HVPN-P2-290..296 · tests: FA, CHAL, REC · DoD: all P2-AC + P2-SLO
 - *Desc:* Run the full-suite retest on a clean fleet, enumerate the discovery-coverage set beyond the reported gates (§11.4.118), and produce the certification report binding every P2-AC/P2-SLO to its green proving item with captured evidence. *Deliverable:* docs/qa/<run-id>/phase2-dod/ report (HTML+PDF+DOCX, §11.4.65/.153). *Acceptance:* all 8 P2-AC + 5 P2-SLO green with evidence paths; the enumerated discovery set surfaces zero release-blockers (or triggers §11.4.129) [04_P2 §10.2, §11.4.40].
-

16. E30 — Governance: docs-chain · items DB · release

HVPN-P2-E30 — Governance / release epic · module: governance

- **HVPN-P2-300 — Phase-2 items DB projection + docs-chain sync.** M(4) · deps: HVPN-P1-150 · tests: UNIT, INT, FA
 - *Desc:* Project this WBS's leaves into docs/.workable_items.db (git-tracked, §11.4.95) via the Phase-1 loader (reused unchanged) and register the docs-chain context so HTML/PDF/DOCX exports + the DB stay in lockstep (§11.4.106/.93/.65/.153). *Deliverable:* loader run + .docs_chain/contexts/wbs-p2.yaml. *Acceptance:* md[?]db byte-identical round-trip (§11.4.93); a leaf edit re-syncs exports out-of-the-box; verify is the gate.
- **HVPN-P2-301 — Phase-2 release tagging + multi-upstream publish.** S(3) · deps: HVPN-P2-297, HVPN-P2-300 · tests: FA · DoD: all P2-AC + P2-SLO
 - *Desc:* Cut the prefixed release tag (<HELIX_RELEASE_PREFIX>-2.0.0-parity from .env else snake_case root dir, §11.4.151) on the main repo **and** every owned submodule with the identical prefix, published to all

upstreams via merge-onto-latest-main (no force-push, §11.4.113).
Deliverable: the tag set + changelog under docs/changelogs/.
Acceptance: `git tag -l '<PREFIX>-2.*'` enumerates the whole release across repos; HVPN-P2-297's certification report is the tag's evidence; the tag is reachable on every remote.

17. Multi-region HA topology (reference diagram)

The production fleet for P2-AC8 / P2-SLO3 / P2-SLO4 (§13). Global control state (HA Postgres + NATS JetStream mesh) is replicated; each region runs stateless coordinators, an edge/relay, and a STUN endpoint; clients pick the nearest edge via Anycast/GeoDNS and an RTT-measured candidate list. Direct-P2P paths bypass the gateway entirely and survive regional failure.

```
graph TD
    subgraph GLOBAL["Global control state (replicated)"]
        PGP["Postgres PRIMARY<br/>(Patroni HA, sync replicas)"]
        NATS["NATS JetStream<br/>cross-region event mesh<br/>(durable subjects = Phase-1 taxonomy 1:1)"]
        PGP -. "regional read replicas" .-> RR1["PG replica EU"]
        PGP -. "regional read replicas" .-> RR2["PG replica US"]
        PGP -. "regional read replicas" .-> RR3["PG replica APAC"]
    end

    subgraph EU["Region EU"]
        CEU["coordinators (N, stateless)<br/>WatchNetworkMap"]
        EEU["edge / helix-relay<br/>+ STUN endpoint<br/>floating IP"]
        RR1 --> CEU
        NATS --- CEU
        CEU --- EEU
    end

    subgraph US["Region US"]
        CUS["coordinators (N, stateless)"]
        EUS["edge / helix-relay<br/>+ STUN endpoint<br/>floating IP"]
        RR2 --> CUS
        NATS --- CUS
        CUS --- EUS
    end

    subgraph APAC["Region APAC"]
        CAP["coordinators (N, stateless)"]
        EAP["edge / helix-relay<br/>+ STUN endpoint<br/>floating IP"]
        RR3 --> CAP
        NATS --- CAP
        CAP --- EAP
    end

    ANY["Anycast IP / GeoDNS<br/>+ RTT-measured gateway list in NetworkMap"]
    CL["Client (Access app)"]
    CN["Connector (advertises LAN CIDRs)"]
    CL -->|"nearest edge"| ANY
```

```

CN -->|"nearest edge"| ANY
ANY -. "routes to" .-> EEU
ANY -. "routes to" .-> EUS
ANY -. "routes to" .-> EAP
CL == "direct P2P (hole-punched, bypasses gateway, survives
region loss)" ==> CN

```

```

classDef g fill:#eef,stroke:#338;
classDef crit fill:#fde,stroke:#c33,stroke-width:2px;
class PGP,NATS crit;

```

Reference deploy artefacts (consumed via the containers submodule, §11.4.76, rootless Podman, §11.4.161):

```

# deploy/quadlets/helix-coordinator.container (Podman quadlet –
stateless, N per region)
[Unit]
Description=HelixVPN stateless coordinator (region=%i)
After=network-online.target
[Container]
Image=localhost/helixvpn/coordinator:2.0.0-parity
Environment=HELIX_REGION=%i HELIX_PG_R0=postgres-replica.%i
HELIX_NATS=nats://nats-mesh:4222
ReadOnly=true
NoNewPrivileges=true
DropCapability=ALL
# coordinator holds NO durable local state – rebuildable cache
(HVPN-P2-270)
[Service]
Restart=always
[Install]
WantedBy=default.target

# deploy/k8s/nats-jetstream.yaml (cross-region event mesh,
durable, P2-SL04)
apiVersion: apps/v1
kind: StatefulSet
metadata: { name: nats-mesh }
spec:
  replicas: 3 # one JetStream peer per
region
  serviceName: nats-mesh
  template:
    spec:
      containers:
        - name: nats
          image: nats:2-alpine
          args: ["-js", "-cluster_name", "helix-mesh", "-routes",
"nats://nats-mesh:6222"]
          # durable subjects mirror the Phase-1 stream taxonomy
          1:1 (transport-only swap, D3)

```

18. DoD work-item traceability matrix

Each release gate must trace to ≥ 1 *implementing* item and ≥ 1 *proving* (test/QA) item. A gate with no green proving item cannot ship (§11.4.123 rock-solid proof).

Gate	Implementing items	Proving items
P2-AC1 full transport set	HVPN-P2-200, 201, 202, 203, 204	HVPN-P2-291, 297
P2-AC2 DAITA efficacy	HVPN-P2-210, 211, 213	HVPN-P2-292, 297
P2-AC3 direct P2P + relay	HVPN-P2-220, 221, 222, 223, 224	HVPN-P2-290, 297
P2-AC4 multi-hop	HVPN-P2-230, 231	HVPN-P2-232, 294
P2-AC5 PQ handshake	HVPN-P2-240, 241, 243	HVPN-P2-242, 293
P2-AC6 desktop apps	HVPN-P2-250, 251, 252, 253	HVPN-P2-297 (per-platform REC)
P2-AC7 GitOps	HVPN-P2-260, 261	HVPN-P2-261, 262
P2-AC8 regional outage survival	HVPN-P2-270, 271, 272, 274	HVPN-P2-295
P2-SLO1 direct-path > 90% / 10 s	HVPN-P2-224	HVPN-P2-290
P2-SLO2 relay→direct < 15 s	HVPN-P2-225	HVPN-P2-290
P2-SLO3 failover p99 < 3 s	HVPN-P2-274	HVPN-P2-295
P2-SLO4 cross-region p99 < 2 s	HVPN-P2-272	HVPN-P2-295, 281
P2-SLO5 PQ adoption > 95%	HVPN-P2-243	HVPN-P2-293, 280

19. Effort roll-up & critical path

Indicative person-day totals (mid-point of the T-shirt range) for sizing, **not** a commitment. Phase-1 deps are prerequisites (excluded).

Epic	Items	~Person-days
E19 Entry-seam gates (prereq)	3	3
E20 Full transport set	7	28
E21 DAITA / maybenot	4	20
E22 Direct P2P + NAT traversal	7	35
E23 Multi-hop	4	19
E24 Post-quantum handshake	5	24
E25 Desktop apps	4	26
E26 Policy-as-code / GitOps	3	14

Epic	Items	~Person-days
E27 HA + multi-region	5	37
E28 Observability	2	8
E29 QA · SLOs · DoD cert	8	41
E30 Governance / release	2	7
Phase-2 total (excl. E19)	51	~259

Critical path (longest dependency chain to DoD certification):

E27(270→271→272→273→274) → E22(220→221→222→223→224→225) → E23(230→231→232) → E29(290→295→297) → E30(301). The two high-risk parallel tracks that converge at E29 but do **not** sit on the networking critical path: **E24 (PQ)** and **E21 (DAITA)** — each is a measure-don't-assert privacy track dispatchable as a disjoint-scope PWU (§11.4.58) off the shipped MVP. **E20/E25/E26** also fan out in parallel. Per §11.4.132 the **correctness-and-downgrade-safety floor** leads every iteration: NAT-traversal honesty (HVPN-P2-222, no false direct), PQ never-weaker-than-classical (HVPN-P2-242), DAITA *measured* efficacy (HVPN-P2-213) — convenience UI (path indicator, pickers, toggles) last.

20. Open decisions surfaced (with recommendation)

Surfaced, not silently resolved (§11.4.66/.101) — each with a recommendation and the item that finalizes it by benchmark/evidence:

- **D1 — Primary obfuscating transport.** Camp A **MASQUE/QUIC** (WG-over-HTTP/3, RFC 9298/9297/9221 — Mullvad's actual mechanism, single Rust impl) vs Camp B **Hysteria2+Salamander** (turnkey, ships faster). *Rec: keep MASQUE/H3 primary* (true parity, one transport stack), evaluate Hysteria2 as an optional ladder rung only on a material reach win — **HVPN-P2-206** decides [SYNTHESIS §3 D1, research-masque, research-hysteria2].
- **D6 — Transport topology.** Single-protocol end-to-end vs **asymmetric per-leg** (Hysteria2/QUIC user[?]gateway, WireGuard gateway[?]networks) [11_MST]. *Rec: stay symmetric MASQUE/H3 user[?]gateway + WG inner* unless **HVPN-P2-206** shows the per-leg split wins on goodput under loss in hostile networks.
- **D3 — Event bus.** Redis Streams (MVP) → **NATS JetStream** (scale). *Rec: swap to NATS at Phase 2* (the EventBus interface is unchanged — transport-only swap) — **HVPN-P2-272** [SYNTHESIS §3 D3].
- **PQ-KEM mechanism.** In-house **ML-KEM PSK exchange** over the authenticated channel vs external **Rosenpass** daemon. *Rec: ship in-house ML-KEM, keep a Rosenpass KemSuite adapter* — **HVPN-P2-244** [04_P2 §5.1].
- **Hybrid hedge depth.** ML-KEM-only PQ secret vs **ML-KEM + Classic McEliece** conservative hybrid. *Rec: ML-KEM as primary; McEliece as an opt-in high-assurance suite* (large keys, setup-only cost) — **HVPN-P2-240/241** [04_P2 §5.1, research-pki_pq_nat].

Decisions whose finalizing item records the evidence land in the §20 decision-log (docs/decisions/) and update the source_refs of the dependent items.

Sources

[04_P2] HelixVPN-Phase2-Parity.md — the full transport set, DAITA/maybenot, direct P2P + NAT traversal + DERP-style relay, multi-hop nested WG, PQ-PSK handshake, Windows/macOS shims, GitOps effect-diff, HA multi-region topology + failover, observability additions, the §10.2 8-criterion DoD + §10.3 SLOs (the spine of this WBS). · [04_ARCH] HelixVPN-Architecture-Refined.md — control/data separation, the Transport trait stance, MASQUE-as-Mullvad-parity (D1), no-logging, ULA /48 + 4via6, repo layout, observability §9 privacy invariant. · [04_P0] HelixVPN-Phase0-Spike.md — the Transport trait, boringtun wrapper, MASQUE/CONNECT-UDP, netns+DPI+netem rig reused by E29. · [04_P1] HelixVPN-Phase1-MVP.md — the seams Phase 2 extends (Coordinator, WatchNetworkMap, TransportPolicy.order, Peer.endpoint, PKI PSK hook, no-log lint, event taxonomy). · [07] 07-phase1-mvp-wbs.md — the WBS format, the §3 items/test_diary DDL (reused unchanged), the §11.4.169 test vocabulary, the id/risk-ordering conventions. · [01]/[02]/[03]/[04]/[00] — the authoritative data-plane / control-plane / client-core / security / scope specs this WBS references, never re-defines. · [11_MST] asymmetric per-leg transport (D6) + Hysteria2 framing. · [07_GMI]/[10_KMI] CGNAT/IPAM and full-ecosystem framing context. · [research-masque] MASQUE / CONNECT-UDP (RFC 9298/9297/9221) WG-over-HTTP/3 as Mullvad's real QUIC mechanism. · [research-hysteria2] Hysteria2 + Salamander obfuscated-QUIC, brutal congestion, turnkey DPI evasion (the D1/D6 alternative). · [research-daita_test] maybenot machines + closed-world website-fingerprinting efficacy methodology (the measure-don't-assert DAITA claim). · [research-pki_pq_nat] ML-KEM (FIPS 203) PSK injection into WG, hybrid-never-PQ-only, Rosenpass alternative, NAT-traversal STUN/hole-punch/relay tradeoffs. · [SYNTHESIS] §§1–9 — the stack floor, the D1–D7 key decisions, ecosystem wiring (containers/helix_qa/challenges/docs_chain), constitution bindings (§11.4.27/.40/.58/.65/.76/.85/.93/.106/.107/.113/.118/.123/.129/.132/.135/.151/.153/.156/.159/.161/.163/.169).